Data Dictionary

E-Commerce Sales Drop Analysis

Important: these are synthetic portfolio-learning datasets, not real company data. Treat hypotheses as questions to test, not established findings.

Business context

An e-commerce business has seen a decline in order volume and revenue in recent months. The analyst must establish when the decline occurred, quantify it, identify which products/customer groups/regions/channels contributed, and recommend evidence-based next steps.

Tables and grain

File	Grain / purpose
orders_raw.csv	Order-line facts. One row represents one product line in an order; order_id is the order identifier, not necessarily unique per line in a real multi-item order. Here the synthetic dataset uses one product line per order ID.
customers_raw.csv	Customer dimension; one row per customer after deduplication.
products_raw.csv	Product dimension; one row per product after deduplication.

customers_clean.csv

Column	Expected type	Definition / usage
customer_id	Text / key	Identifier used to distinguish or join records; preserve as text.
customer_name	Text / category	Descriptive dimension used for grouping, filtering, or segmentation.
city	Text / category	Descriptive dimension used for grouping, filtering, or segmentation.
region	Text / category	Descriptive dimension used for grouping, filtering, or segmentation.
customer_segment	Text / category	Descriptive dimension used for grouping, filtering, or segmentation.
signup_date	Date / datetime	Time field used for filtering, trends, cohort assignment, or event sequencing.
acquisition_channel	Text / category	Descriptive dimension used for grouping, filtering, or segmentation.

orders_clean.csv

Column	Expected type	Definition / usage
order_id	Text / key	Identifier used to distinguish or join records; preserve as text.
order_date	Date / datetime	Time field used for filtering, trends, cohort assignment, or event sequencing.
customer_id	Text / key	Identifier used to distinguish or join records; preserve as text.
product_id	Text / key	Identifier used to distinguish or join records; preserve as text.
quantity	Integer	Count, elapsed-period, binary outcome, or data-quality flag; confirm meaning before aggregation.
unit_price	Numeric	Measure used in calculations; confirm units and formula before interpreting.
discount_pct	Numeric	Measure used in calculations; confirm units and formula before interpreting.
revenue	Numeric	Measure used in calculations; confirm units and formula before interpreting.
estimated_cost	Numeric	Measure used in calculations; confirm units and formula before interpreting.

Column	Expected type	Definition / usage
sales_channel	Text / category	Descriptive dimension used for grouping, filtering, or segmentation.
order_status	Text / category	Descriptive dimension used for grouping, filtering, or segmentation.
month	Text (YYYY-MM)	Calendar month derived or assigned for trend/cohort analysis.
gross_sales	Numeric	Measure used in calculations; confirm units and formula before interpreting.
discount_amount	Numeric	Measure used in calculations; confirm units and formula before interpreting.
profit_estimate	Numeric	Measure used in calculations; confirm units and formula before interpreting.

products_clean.csv

Column	Expected type	Definition / usage
product_id	Text / key	Identifier used to distinguish or join records; preserve as text.
product_name	Text / category	Descriptive dimension used for grouping, filtering, or segmentation.
category	Text / category	Descriptive dimension used for grouping, filtering, or segmentation.
list_price	Numeric	Measure used in calculations; confirm units and formula before interpreting.
unit_cost	Numeric	Measure used in calculations; confirm units and formula before interpreting.
supplier_id	Text / key	Identifier used to distinguish or join records; preserve as text.

Relationships and metric cautions

• orders.customer_id → customers.customer_id (many order lines can map to one customer).

• orders.product_id → products.product_id (many order lines can map to one product).

• Orders have one product line per order ID in this synthetic dataset; still verify grain before counting orders.

• Profit is an estimate based on the supplied estimated_cost, not audited accounting profit.